

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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Aim:

To write a Prolog program to implement **pattern matching** between two lists.

Objective:

To check whether a given pattern matches a sequence of elements using Prolog's **unification and recursion**.

Algorithm:

1. ☐ Start the program.
2. ☐ Read the pattern and input list.
3. ☐ Compare the first element of the pattern with the first element of the input.
4. ☐ If they match, recursively compare the remaining elements.
5. ☐ If both lists become empty, the pattern matches successfully.
6. ☐ If any element does not match, return false.
7. ☐ Stop.

Prolog Program:

% Pattern Matching Program

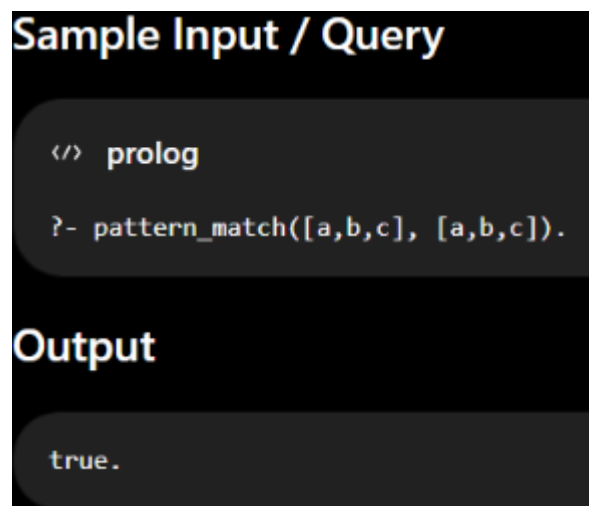
```
pattern_match([], []).
```

```
pattern_match([H1|T1], [H2|T2]) :-
```

```
    H1 = H2,
```

```
    pattern_match(T1, T2)..
```

Output

A screenshot of a Prolog IDE interface. It features a dark background with light-colored text. At the top, the text "Sample Input / Query" is displayed in a yellow font. Below this, a code editor shows two lines of Prolog code: a comment line starting with "</> prolog" and a query line "?- pattern_match([a,b,c], [a,b,c]).". The query line is highlighted with a light blue background. Below the code editor, the word "Output" is written in a light blue font. Underneath, a text box displays the result "true." in a light blue font.

```
Sample Input / Query
```

```
</> prolog
```

```
?- pattern_match([a,b,c], [a,b,c]).
```

```
Output
```

```
true.
```

Result:

Thus, the Prolog program successfully implements **pattern matching using unification and recursion**.